

CPU Central Processing Unit

CPU Architecture

x86 / x64

ARM

Desktop & Server CPU

* PGA (Pin Grid Array) Sockets (holes)

* LGA socket (Land Grid Array) (pads)

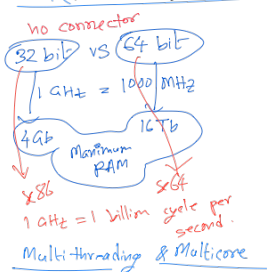
Intel created

Server CPU Sockets

Multi Socket Motherboards

Mobile CPU Sockets

BGA (Ball Grid Array) Socket

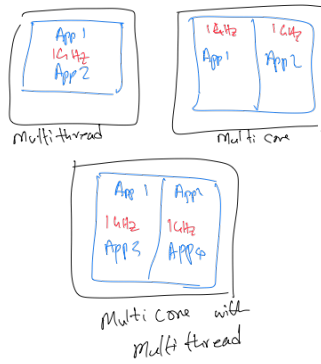


Multi-threading & Multicore

Hyper-threading

SMT (Simultaneous multi-threading)

Multicore:



CPU Technology

Virtualization Support

↳ Host (VMs) Virtual Machines

↳ enabled/disabled

↳ BIOS / UEFI

Inter Virtualization Technology

(Intel VT)

Intel VT-d

AMD (AMD-V)

Secure Virtual Machines

Expansion Cards

* Sound Card

* Video Card → PCIe card

* Capture Card

* NIC

GPU
SLI technology

Motherboard

Read the manual

PCIe

(Peripheral Computer Interconnect)

latest & fastest slot

PCIe Express

latest & fastest slot

different sizes → internal expansion

Mini PCIe

Matrix Power Connector

↳ 5V & 12V

↳ standard connector

SATA Drive Power Connector

↳ 5V, 12V & 3.3V

PCIe Power Connector

6 pin → 3 rails → 12V

8 pin → 4 rails → 12V

Video card

ATX connector

↳ 5V, 12V & 3.3V

↳ to motherboard

↳ 20 pin & 24 pin

ATX 12V

↳ 4 pin → P4 connector

↳ To motherboard CPU

BIOS & UEFI

Computer Firmware

Basic Input Output System

Unified Extensible Firmware

ROM (Read Only Memory) chip

store → BIOS / UEFI → new system

UEFI → Secure Boot

Faster startup

mouse & keyboard

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CMOS (Complementary Metal-Oxide-Semiconductor)

CMOS Setup Utility

BIOS Setup Utility / UEFI Setup Utility

RTC (Real Time Clock)

Real Time Password

located → Volatile Memory

(CR2032) coin batteries

Clear CMOS (Switch / Jumper)

CMOS settings

1. Boot device Priority
2. Boot Time Password
3. Enable / Disable access to Boot
4. Wake-on-LAN (WoL)
5. Virtualization Support
6. CPU multipliers & voltages
7. RAM timing and voltages

Power Supply Unit (PSU)

Watts → 500 Watt → standard PSU

Non Modular PSUs

Modular PSUs

Redundant Power Supplies

(ATX vs SFX) PSU

ATX vs SFX motherboards

mini ITX or smaller form factor

Security

* User Authentication

* Badging

* Audit logs

* Secured Prints

ADF (Automatic Document Feeder)

Flatbed Scanner

Finishes

↳ Duplexers

↳ Collator / Separator

↳ Stapler

↳ Binder

↳ Paper Trays

Laser Printer Maintenance kit

↳ Fuser Unit

↳ Transfer Roller

↳ Paper Feed Roller / Separation Roller

↳ Pickup Roller

Clean & Calibrate

W4 5 to 7 Questions will come

Laser Printer

fixe powered

Print Process

ink onto paper by using

very high heat to melt

toner on to the page.

→ advantage:-

* Fastest Printer

* Lowest cost per page (COP)

for Black & White

→ disadvantage:-

High cost → color print

→ Maintenance needed